Course Title	COMPUTER ARCHITECTURE LAB		
Course Code:	CSC 223-L	**Semester:**	Fall 2026
Credit Hours:	1 (Lab), 3 contact hours per week (course: 2+1)	**Prerequisite(s):**	CSC 222
Instructor:	Faiza Ali (Junior Lecturer)	**Class:**	BS (Computer Science)
Office:	A8	**E-mail:**	faiza.ali@iqraisb.edu.pk
Lab Days:	As per timetable (one 3-hour session per week)	**Consultation Hours:**	As announced on LMS
Lab Room:	Computer Lab (as per timetable)	**Knowledge Group:**	Architecture and Organization (ACM KA: AR)

Course Description:

This lab supports the theory course *Computer Architecture* (CSC 222/223). Students explore the instruction set architecture (ISA) of a CISC machine (Intel x86) and a RISC machine (MIPS) through assembly language programming, and simulate key architectural mechanisms. The lab covers:

- x86 ISA, registers, flags, memory segmentation and byte ordering using the emu8086 emulator
- Assembly language programming: data transfer, arithmetic, branching, loops, arrays, procedures and macros
- MIPS ISA, SPIM system calls, procedures and the run-time stack using the QtSPIM simulator
- Vector operations and loop unrolling, cache read/write behaviour and virtual-to-physical address translation

Course Objectives:

✓ To develop a clear understanding of instruction set architecture by programming x86 (CISC) and MIPS (RISC) processors at the assembly level.

✓ To enable students to analyse how high-level constructs (decisions, loops, arrays, procedures) are implemented with registers, memory and the stack.

✓ To give hands-on experience with simulators (emu8086, QtSPIM) for observing program execution, memory hierarchy (cache) and virtual memory behaviour.

Course Learning Outcomes (CLOs):

	At the end of the course, the students will be able to:	GA	BT Level*
1	Apply concepts of instruction set architecture and assembly language programming to develop solutions for x86 and MIPS systems.	GA2	P-3
2	Analyze and implement system-level operations, such as procedures, macros, and memory management, using assembly language and simulators.	GA6	A-3

* BT = Bloom's Taxonomy, C = Cognitive domain, P = Psychomotor domain, A = Affective domain. CLO weight: CLO1 = 65%, CLO2 = 35%. SDG: 4 (Quality Education).

Mapping of CLOs to GAs		
GAs / CLOs	**CLO1**	**CLO2**
GA1 (Academic Education)		
GA2 (Knowledge for Solving Computing Problems)	**X**	
GA3 (Problem Analysis)		
GA4 (Design/Development of Solutions)		
GA5 (Modern Tool Usage)		
GA6 (Individual and Team Work)		**X**

GA1 (Academic Education) is attained automatically on successful completion of the course.

Week No.	Topic / Contents	Lab No.	CLO
Part I: x86 Assembly Language Programming (emu8086)			
Week 1	Exploring Instruction Set Architecture (ISA) of x86 Machines • Registers, flags, memory segmentation, byte ordering (little/big endian); emu8086 environment	1	1-2
Week 2	Learning to program in Assembly Language of x86 Machines • Program structure, directives, data definition, assembling, single-stepping, listing file	2	1-2
Week 3	Using MACROS for Input / Output and Data Conversion • INT 21h services, I/O macros, ASCII ↔ binary/decimal/hex conversion	3	1-2
Week 4	Using x86 Data Transfer and Arithmetic Instructions • MOV, XCHG, LEA, PUSH/POP, XLAT; ADD, SUB, INC, DEC, NEG, MUL, DIV and flags	4	1-2
Week 5	Implementing Branching in x86 Assembly Language • CMP, JMP, signed/unsigned conditional jumps, if-else structures	5	1-2
Week 6	Implementation of Loop Structures in x86 Assembly Language • LOOP, counter-, sentinel- and condition-controlled loops, nested loops	6	1-2
Week 7	Open Ended Lab / Project Assigned	–	1-2
Mid Semester Examination (Week 8)			
Week 9	Array Processing in x86 Assembly Language • Indexed/based addressing, sum, max/min, search, bubble sort	7	1-2
Week 10	Development of Procedures and Macros in x86 Assembly Language • CALL/RET, stack frames, parameter passing, recursion, macros with parameters	8	1-2
Part II: MIPS Assembly Language and Memory System (QtSPIM)			
Week 11	Familiarization with SPIM – a MIPS simulator • MIPS registers, QtSPIM interface, R/I-type instructions, load/store	9	1-2
Week 12	SPIM console, system calls and Procedures in MIPS Assembly • syscall services, jal/jr, argument/return registers, stack frames, recursion	10	1-2
Week 13	Vector operations in MIPS Assembly and Loop Unrolling • Vector add, dot product, unrolling and its effect on instruction count	11	1-2

Week No.	Topic / Contents	Lab No.	CLO
Week 14	Simulating Cache Read/Write • Direct-mapped cache: tag/index/offset, hits, misses, conflict misses (QtSPIM simulation)	12	1-2
Week 15	Learning Address Translation in a Virtual Memory System • Pages, page table, VPN/offset, page faults (QtSPIM simulation)	13	1-2
Week 16	Open Ended Lab / Project Assessment	–	1-2
Week 17	Final Examination	–	1-2

Lab Policies	
Software:	emu8086 (8086 microprocessor emulator) for Labs 1–8; QtSPIM (MIPS32 simulator) for Labs 9–13. Both run on the lab computers; students are advised to install them on their own laptops.
Lab Task Policy:	Each lab task is completed and demonstrated in the same session and graded on the lab rubrics. Programs and screenshots are uploaded to the LMS before the deadline. Late submissions will not be entertained.
Quiz Policy:	There are no separate quizzes in this lab (as per course outline); understanding is assessed through lab tasks, viva and examinations.
Open Ended Lab:	Assigned in Week 7 and carried out in groups of 2–3 students; assessed through demonstration, viva and report in Week 16.
Attendance:	As per university policy. Missed labs must be completed before the next session.
Plagiarism:	Collaboration is encouraged but each student must write and explain his/her own code. Copied programs (from peers, the internet or AI tools) receive zero marks for all rubrics of that task and may be referred to the HoD for appropriate action.

Prerequisites:

CSC 222 (Computer Architecture theory). Familiarity with number systems (binary, hexadecimal, two's complement) and basic programming is expected.

Course Textbook / Reference Books			
S No	**Book Title**	**Author(s)**	**Edition / Publisher**
1	Computer Organization and Architecture: Designing for Performance	William Stallings	11th Ed., 2019, Pearson
2	Computer System Architecture	M. Morris Mano	3rd Ed., 2017, Pearson Education
3	Computer Organization and Design: The Hardware/Software Interface (MIPS Edition)	D. A. Patterson, J. L. Hennessy	Morgan Kaufmann (reference for Labs 9–13)
4	Assembly Language Programming and Organization of the IBM PC	Ytha Yu, Charles Marut	McGraw-Hill (reference for Labs 1–8)

Grading Policy

Assessment				
Assessments / CLOs	**CLO1**	**CLO2**	**Total**	**Week**
Lab Manual (lab tasks): 20%	10%	10%	20%	1–15
Mid Semester Exam (MSE): 25%	25%	–	25%	8
Project / OEL: 15%	–	15%	15%	7–16
End Semester Exam (ESE): 40%	30%	10%	40%	17
Total	**65%**	**35%**	**100%**	

Weights follow Section 6 (CLO Outcome-Based Assessment) of the course outline. Every CLO is assessed at least three times.